Inward Resolutions of the Fermi Paradox: A Critical Review of Migration Down Thermodynamic Gradients

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Abstract

Most catalogued resolutions of the Fermi paradox modify one of three things: the abundance of technological life, its longevity, or its visibility. A smaller family modifies its direction: these are the inward-migration hypotheses, which hold that mature technological intelligence migrates down thermodynamic gradients, toward denser, serially faster, colder and more computationally efficient configurations of matter, and that the observed silence of the sky is the external appearance of that migration. The family comprises six distinguishable proposals, built on a six-decade precursor lineage from Dyson's eternal-computation bound to Bradbury's Matrioshka-brain engineering: the migration hypothesis of 'Cirkovi'c and Bradbury, Smart's transcension hypothesis, Vidal's stellivore interpretation, the aestivation hypothesis of Sandberg, Armstrong and 'Cirkovi'c, black-hole computing proposals from Inoue and Yokoo through Dvali and Osmanov, and the recent Macro Transcension Hypothesis. They share one load-bearing premise: that the thermodynamics of computation, rather than expansion, reproduction, or communication, is the correct lens for predicting the behaviour of the oldest intelligence.

The family has not previously been reviewed as such, nor its mutual inconsistencies catalogued or its falsifiability graded. We reconstruct the family tree and its intellectual debts; restate the unifying physics (Landauer's principle, the Margolus–Levitin bound, Bekenstein–Hawking entropy, and the temperature hierarchy of available entropy sinks) with explicit numbers; construct a comparative matrix of assumptions, energy logics, predicted observables, standing objections, and observational status for each member; and grade each against six falsifiability criteria. Against the sociological solutions that predict the same silence from different premises (zoo, dark-forest, and sustainability), the inward family's distinguishing virtue is residue: thermodynamic optimization leaves dynamical and high-energy traces that fear and ethics do not. Open problems are stated as research questions: goal stability over 10^8-year horizons, migration economics under Bostrom-type opportunity costs, the incomplete-compliance gap, and population-level consistency with grabby-aliens selection effects. We close with the instrument-matched tests now feasible for each hypothesis, anchored by the first dedicated globular-cluster technosignature surveys and the multi-messenger campaign proposed for Omega Centauri.

Keywords: Fermi paradox; SETI; technosignatures; transcension; aestivation; black-hole computing; Landauer limit; Dyson spheres; dark forest; intermediate-mass black holes

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1 Introduction

1.1 The expansionist premise

The Fermi paradox in its canonical form (Hart, 1975; Tipler, 1980; Brin, 1983; Webb, 2015; Ćirković, 2018; Forgan, 2019b) is an argument from absence built on a premise about direction: technological civilizations, if they endure, grow outward. The premise has quantitative teeth. Self-reproducing probes colonize the Galaxy in 10^6–10^8 years (Tipler, 1980; Freitas, 1980; Freitas and Gilbreath, 1982); a single mature civilization can seed every galaxy in the reachable universe, out to ∼1200 Mpc, for a fraction of one star's output (Armstrong and Sandberg, 2013); and expansion, once begun anywhere, is self-amplifying, so that even rare expansionists should dominate the visible universe on cosmological timescales (Olson, 2015; Hanson et al., 2021). On the expansionist premise, the silence is genuinely paradoxical, and the escape routes (life is rare, life dies young, life hides) are each independently uncomfortable.

Sixty years of searching have sharpened the absence. Radio and optical SETI have found no beacons (Tarter, 2001; Enriquez et al., 2017; Huang et al., 2026b). Waste-heat surveys, the most general possible search since the second law guarantees that energy harvesting at scale produces mid-infrared excess (Dyson, 1960), have found no Kardashev Type II/III candidates: not in IRAS all-sky data (Carrigan, 2009), not among 10^5 galaxies surveyed with WISE in the $\hat{\mathrm{G}}$ program (Wright et al., 2014b,a; Griffith et al., 2015), not in radio–infrared correlation follow-up of the best $\hat{\mathrm{G}}$ candidates (Garrett, 2015), not in Tully–Fisher residuals of disk galaxies (Zackrisson et al., 2015), not among nearly complete Dyson spheres searchable with Gaia photometry (Zackrisson et al., 2018), and not as star-fed Type III engineering in any surveyed population (Annis, 1999). The most recent chapter is instructive on both counts. Project Hephaistos combed Gaia, 2MASS, and WISE photometry for partial Dyson spheres, first in $\sim 2.7 \times 10^5$ nearby stars (Suazo et al., 2022) and then in five million, reporting seven M-dwarf candidates with unexplained mid-infrared excess (Suazo et al., 2024); follow-up work identified radio counterparts offset from the stars (Ren et al., 2024), showed the excesses are consistent with chance alignments of dust-obscured background galaxies (Blain, 2024), and resolved the radio-brightest candidate's counterpart into a background radio galaxy (Ren et al., 2025). A systematic archival-diagnostics pass across the nine candidates found significant background contamination in two and suggestive-to-marginal evidence of it in five more, leaving two of the nine (candidates D and I) without an identified contaminant (Ren et al., 2026); targeted follow-up spectroscopy and photometry of the original seven plus three related sources likewise found no stellar irregularity but no confirmed Dyson-sphere signature either, with two objects newly resolved as superpositions on red background galaxies (Korn et al., 2026). Two features of the episode matter here: the most modern candidate search has narrowed toward null without fully closing it, with two candidates still open pending dedicated follow-up, and the contamination fight demonstrates the confusion-limited systematics that any infrared search for the structures reviewed here will face (Section 7, Table 4). What sixty years of silence actually excludes must also be calibrated against our own detectability: an "Earth Detecting Earth" accounting shows present-day instruments would detect present-day Earth's strongest technosignatures only out to modest distances (Sheikh et al., 2025), so the surveys above bound loud engineering at scale, not quiet technology. Within those limits, the instruments built on the expansionist premise have steadily undermined it.

1.2 The inward alternative

The alternative reviewed here inverts the direction. Its claim, in the most general form: *for an intelligence that optimizes computation rather than territory, the gradient of value in the universe points inward, toward higher densities, faster clock rates, colder entropy sinks, and more efficient mass–energy conversion, and*

the visible universe is therefore not where mature intelligence lives. Different members of the family locate the endpoint differently (planet-scale "inner space," engineered Matrioshka shells, the cold far future, stellar interiors, or existing black holes), and differ on whether the migration is metaphorical, developmental, or literal relocation. All share the premise that the thermodynamics of computation, rather than the cost of transportation, sets the direction of mature development.

This family has not previously been reviewed as a family. The nearest antecedent is the Dyson Minds workshop report (Curtis et al., 2026), which develops observational strategies for black-hole-hosted intelligence but confines itself to supermassive hosts and does not catalogue the family's internal contradictions or grade its falsifiability. The members themselves are scattered across six decades, three disciplines, and widely varying registers, from *Reviews of Modern Physics* (Dyson, 1979) to unpublished engineering manuscripts (Bradbury, 1999) to philosophy journals (Bostrom, 2003), and they are routinely conflated: transcension is glossed as aestivation, aestivation as hiding, black-hole computing as Dyson engineering. The conflations matter because the proposals make different and partly contradictory predictions, and because their falsifiability ranges from genuinely testable to unfalsifiable in principle. A review that sorts them is overdue on ordinary scholarly grounds, and newly practical for an observational one: the instruments now arriving (surveyed in Section 7) make several of the family's predictions testable for the first time, and the first dedicated tests are being proposed and conducted (Huang et al., 2026b; Swanson, 2026d).

1.3 Scope, method, and disclosure

We review proposals that satisfy two criteria: (i) they explain the Fermi observation primarily through the direction of mature technological development rather than through rarity, doom, or deliberate concealment; and (ii) they ground that direction in physics, specifically the thermodynamics and physical limits of computation, rather than in strategic or sociological postulates. Criterion (ii) excludes the zoo hypothesis (Ball, 1973), the dark-forest argument (Brin, 1983; Yu, 2015; Forgan, 2017), and the sustainability solution (Haqq-Misra and Baum, 2009), whose silence-mechanisms rest on game-theoretic or policy premises rather than physical limits; these are treated in Section 5 as the family's nearest competitors, because they predict the same silence from different premises and the comparison is instructive.

Disclosure. The author proposed the most recent member of this family, the Macro Transcension Hypothesis (Swanson, 2026c, hereafter Paper A), and the observational campaign attached to it (Swanson, 2026d, hereafter Paper C). A review by a participant carries an obvious hazard. We mitigate it mechanically: every hypothesis including the author's is run through the same comparative matrix (Table 2) and the same falsifiability scorecard (Table 3) with criteria stated in advance (five of the six; the sixth, C6, was added in revision at referee prompting and is disclosed as such in Section 4.2); the scorecard penalizes the failure modes a partisan would excuse; and the standing objections to the MTH are catalogued at the same depth as those to its relatives. Readers should apply their own discount.

1.4 Structure

Section 2 states the unifying physics with numbers. Section 3 reconstructs the family member by member. Section 4 presents the comparative matrix and the falsifiability scorecard. Section 5 contrasts the family with the sociological silence-solutions. Section 6 states the open problems. Section 7 maps the observational program, and Section 8 concludes.

2 The unifying physics, with numbers

Epistemic status: This section is established physics, with the contested edges flagged. None of it presupposes extraterrestrial intelligence; it establishes only what an arbitrarily capable computation-optimizer would care about. Detailed derivations for the black-hole entries appear in Paper A, Section 4; here we need only the magnitudes.

2.1 The price of a bit: Landauer's principle

Erasing one bit of information in an environment at temperature T dissipates at least $k_B T \ln 2$ (Landauer, 1961): 2.9×10^{-21} J at room temperature, 2.6×10^{-23} J against the cosmic microwave background (CMB) at $T_\gamma = 2.725$ K. The bound is no longer merely theoretical: it has been verified experimentally at the single-bit level (Bérut et al., 2012), and its statistical-mechanical foundations, though subtle, have survived modern scrutiny (Parrondo et al., 2015; Wolpert, 2019). Logically reversible computation evades the bound for computation itself (Bennett, 1973; Fredkin and Toffoli, 1982; Bennett, 1982; Frank, 2002, 2017); but error correction, measurement, and any irreversible output commit the computer to erasure, so every long-running physical computation has a T-proportional operating cost, set by the temperature of the sink receiving the waste entropy. That dependence is the shared starting point of every hypothesis reviewed here.

2.2 The speed and capacity of matter

Two further bounds fix the value of density. The Margolus–Levitin theorem caps state transitions at $2E/\pi\hbar$, i.e. $\approx 6 \times 10^{33}$ operations per second per joule of invested energy (Margolus and Levitin, 1998); Lloyd's "ultimate laptop" analysis applies this to the full rest energy of one kilogram of matter, $E = mc^2$, giving $\sim 5 \times 10^{50}$ ops s^{-1}, a total that compression redistributes between memory capacity and serial depth and does not raise at any density (Lloyd, 2000): compression toward black-hole density converts a slow, parallel, memory-rich machine into a fast, serial one. Information capacity obeys the Bekenstein bound (Bekenstein, 1981) and is saturated only by black holes, whose Bekenstein–Hawking entropy (Bekenstein, 1973; Hawking, 1975) corresponds to $\sim 10^{86}$ bits for a $2 \times 10^4\ M_\odot$ object, exceeding by tens of orders of magnitude any archive constructible from ordinary matter of the same mass. The holographic generalization (Bousso, 2002) makes the point structural: the densest possible information storage in nature is a horizon. Both bounds thus favour compression, and black holes are the limiting case for both serial depth and storage.

2.3 The sink hierarchy

The third ingredient, and the one on which the family members genuinely diverge, is the menu of available entropy sinks (Figure 1). Today's ambient sink is the CMB at 2.725 K. The far future offers better: as the universe expands the CMB cools toward the de Sitter horizon temperature, $T_{\rm dS} \approx 2.7 \times 10^{-30}$ K, the ultimate environmental floor (Krauss and Starkman, 2000); waiting converts each joule into up to $\sim 10^{30}$ times more erasures (Sandberg et al., 2016), a strategy anticipated in Dyson's analysis of eternal computation in an open universe (Dyson, 1979). But the present epoch already contains sinks colder than the far-future CMB of any reachable era: black-hole horizons. The Hawking temperature $T_{\rm H} \approx 6.2 \times 10^{-8}\ (M_\odot/M)$ K is 3×10^{-12} K for a $2 \times 10^4\ M_\odot$ IMBH and 1.4×10^{-14} K for Sgr A* (taking $M_{\rm Sgr\,A^*} \approx 4.3 \times 10^6\ M_\odot$; GRAVITY Collaboration 2022): twelve to fourteen orders of magnitude colder than the CMB, available in the present epoch with no waiting required (relative to a 300 K biosphere,

the corresponding minimum erasure costs are fourteen to sixteen orders of magnitude smaller). Entropy dumped across a horizon increases its area in accordance with the generalized second law (Bekenstein, 1974), at marginal cost set by the horizon's effective temperature rather than the sky's.

A contested step must be flagged here, because much of the family leans on it. Treating T_{H} as the achievable marginal erasure cost assumes the computer can dump entropy at the horizon temperature; but a physical computer must radiate its waste heat at its own operating temperature toward the horizon, and any realistic machine sitting outside the hole operates far above T_{H}. The effective sink temperature is then set by the radiator, not the horizon, and the advertised 10^{12+} gain shrinks accordingly. This is the same class of objection that Bennett et al. (2019) raised against aestivation's far-future accounting. One member of the family has answered it: Paper A (§4.2) proposes encoding waste entropy on long-wavelength carrier photons beamed into the horizon's capture cone, a delivery-cost circumvention rather than a material radiator approaching T_{H}. The reply also settles the vocabulary of a related objection from energy conservation, since the mainstream position holds that Dyson-scale computation must reradiate nearly all the energy it absorbs as mid-infrared waste heat (Curtis et al., 2026): under carrier delivery the energy crosses the horizon together with the entropy, so only the fraction of carriers that miss the capture cone is reradiated, at the swarm's operating temperature. The disputed quantity is accordingly a fraction of *energy* delivered (the f_{sink} of Paper A §4.2, Swanson 2026c, quantified further by Swanson 2026b), and the disagreement with Curtis et al. (2026) is a clean, quantitative one about how large that fraction can be made. We grade Paper A's reply partially adequate (⊙, open): the delivery mechanism is exhibited, and Paper E prices the capture-cone, beaming, and étendue geometry directly, but the re-interception term, the probability that carriers are caught and re-thermalized within the swarm itself, is left an estimate rather than a full calculation, its bounded and signed influence on Paper E's own adjudication assessed by sensitivity analysis rather than derived from first principles (Swanson, 2026b); until re-interception is computed directly the objection is open on that ground, and Tables 2 and 3 carry it in that form. With that caveat registered, the sink hierarchy remains the family's organizing axis: each hypothesis amounts to a claim about which sink mature intelligence uses and when.

2.4 Scope of the gradient argument

The gradient argument establishes a conditional: if an agent's terminal values reduce to maximizing long-term computation (or anything computation can purchase), then its resource allocation should flow inward, toward density, spin, and cold sinks. The premise is substantive and contestable: it is the same optimization-pressure postulate adopted, in different forms, by every member of the family (Smart, 2012; Sandberg et al., 2016; Vidal, 2014), and we treat objections to it in Section 6. What the gradient argument does not establish is uniqueness of endpoint: the family's internal disagreements (Section 3) are disagreements about where on the gradient the optimum sits, and several members locate it in places that the physics above does not obviously prefer. Nor does the physics of black holes by itself select the inward reading: Gould (2022) presses the same objects into expansionist service, proposing a gravitational slingshot from a stellar-mass black hole orbiting Sgr A* that launches robotic craft toward M31 at 0.1 c, and arguing that feasible intergalactic colonization would only tighten the Fermi paradox. The comparative matrix exists to make those disagreements explicit.

3 The family, member by member

Figure 2 arranges the lineage; this section treats each member critically. We use a fixed template: *claim*, *mechanism*, *predicted appearance of the sky*, *standing objections*, *current observational status*.

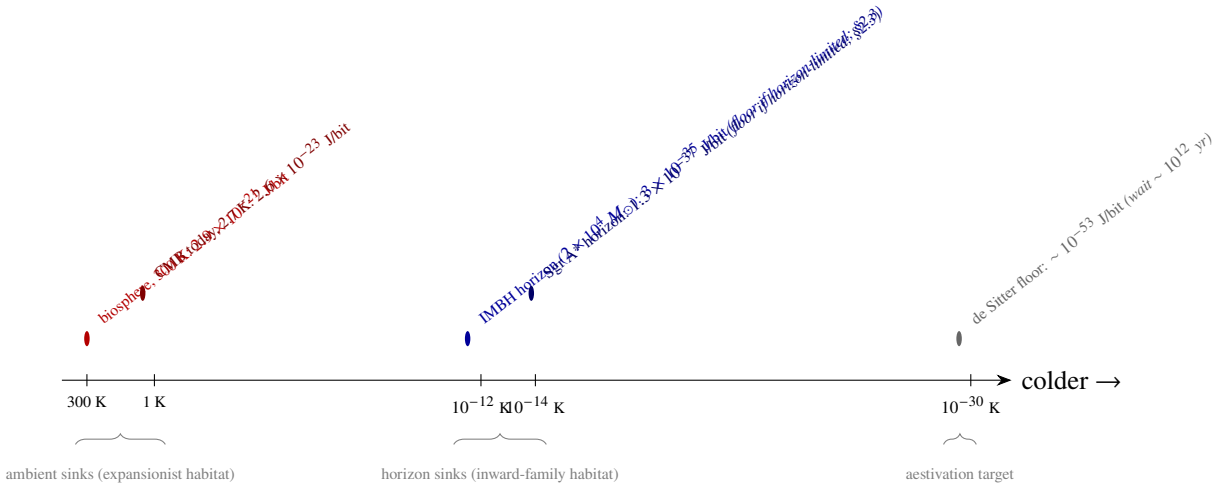


Figure 1: The entropy-sink hierarchy: minimum erasure cost $k_B T \ln 2$ versus sink temperature (horizontal coordinate $-\log_{10}(T/\mathrm{K})$, temperature decreasing rightward). The horizon-sink entries are floors, realized only if entropy can be delivered at the horizon temperature; Paper A's carrier-delivery proposal addresses the delivery step but leaves re-interception within the swarm unpriced (Section 2.3). The inward family's central observation is that horizon sinks colder than any achievable future CMB exist *in the present epoch*; the aestivation strategy of waiting for the de Sitter floor (Sandberg et al., 2016) buys ∼18 further decades of efficiency, measured from this figure's IMBH horizon-sink entry (3×10^{-35} J/bit) down to the de Sitter floor rather than from today's CMB at the price of $\sim 10^{12}$ years of cooling (69 e-folds of expansion at $H^{-1} \approx 1.5 \times 10^{10}$ yr) and the free-energy losses identified by Bennett et al. (2019).

3.1 Precursors: Dyson, Barrow, Bradbury, Bostrom

Four threads predate any explicit inward Fermi solution. Dyson (1960) made energy harvesting at stellar scale thinkable and, crucially for everything after, made waste heat the universal observable. Dyson (1979) then showed that computation in an open, cooling universe is in principle unbounded if clock rates slow with the ambient temperature: the first rigorous treatment of computation as the long-run currency of intelligence, later sharpened against Λ-dominated expansion, which imposes a finite total (Krauss and Starkman, 2000). Barrow (1998) formalized the inward axis itself, ranking civilizations by mastery of ever-smaller scales as a complement to the Kardashev energy scale (Kardashev, 1964). Bradbury (1999) engineered the first concrete inward artifact, the Matrioshka brain, and discovered its binding constraint: nested shells are limited by radiator area and sink temperature rather than by energy supply, the constraint that later black-hole proposals dissolve. The contemporaneous peer-visible quantitative treatment of those limits is Sandberg (1999), which derives the heat-dissipation and internal-connectivity constraints on planet-scale computing superobjects; the two analyses are cross-fertilizing rather than ancestral, Bradbury's manuscript circulating while Sandberg's appeared in print. Two further objections now stand against the Bradbury lineage as engineering: swarm-element stability requires active stationkeeping against radiation pressure and perturbations (Wright, 2020), and Lacki (2025) shows that Kardashev II megaswarms are destroyed by collisional cascades on timescales short compared to the occupation times the hypotheses require, unless continuously policed. Any inward proposal that routes through Matrioshka-style construction inherits both. Bostrom (2003), from philosophy, supplied the economic frame: every century of delayed colonization forgoes a fixed, irrecoverable harvest of free energy, an argument that cuts against waiting strategies and therefore sits in productive tension with aestivation, as Bennett et al. (2019) would later make rigorous.

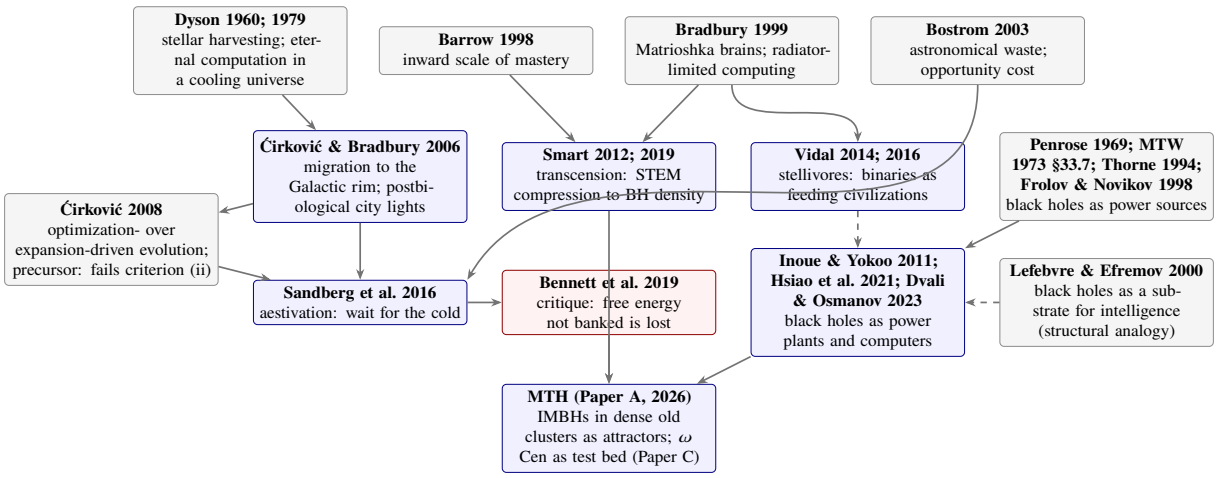


Figure 2: The inward-family lineage. Grey: precursors that supplied components without meeting both membership criteria of Section 1; the pre-2006 black-hole-power thread (Penrose through Frolov & Novikov; Section 3.6), Ćirković (2008) (graded a precursor on criterion (ii); Section 3.1), and Lefebvre and Efremov (2000) (the earliest black-holes-as-substrate source found; Section 3.1) enter here. Blue: family members reviewed in Section 3. Red: the pivotal critique. Solid arrows indicate intellectual descent documented by citation or explicit response; dashed arrows are thematic. The stellivore → black-hole-computing arrow is thematic rather than a documented descent: Inoue and Yokoo (2011) (December 2011) predates Vidal (2014) and neither cites the other. Vidal (2011) (April 2011), an independent early statement of the black-holes-as-attractor thesis, is placed in the black-hole-infrastructure thread rather than the stellivore thread on content grounds. The earliest statement found predates it by a decade and sits outside the cited literature: Lefebvre and Efremov (2000), published in Russian in 2000, proposed black holes as a physical substrate for intelligent beings on grounds of structural analogy, arguing four parallels with the brain–psyche system (complementary internal and external descriptions, an isomorphism between the inner structure of a rotating hole and their model of reflexive cognition, a shared thermodynamic facet, and holography) rather than any energy logic; it enters the figure as a grey node feeding the black-hole-infrastructure thread thematically.

One further thread postdates the family's founding paper yet belongs here with the precursors. Ćirković (2008) argues, with two toy models of postbiological evolution, that the generic developmental pathway of advanced civilizations is optimization-driven rather than expansion-driven, and that the spatially compact option follows; it is the closest prior statement of this review's organizing thesis, written by a co-author of the family's founding paper. We grade it a precursor rather than a seventh member because its grounding is evolutionary and economic: it supplies the direction argument without the thermodynamic mechanism, and so fails membership criterion (ii) of Section 1. Figure 2 records it as a grey precursor node; the family count, matrix, and scorecard are unchanged by it.

The grey-literature sweep that prepared this review found one precursor older than all four threads above. Lefebvre and Efremov (2000), published in Russian in the popular-science journal *Zemlya i Vselennaya* with an English version on arXiv, hypothesize that black holes may serve as a physical substratum for intelligent beings, on the strength of four parallels between the brain–psyche system and black holes: complementary descriptions by internal and external observers, an isomorphism between the inner structure of a Kerr hole and the subjective domain in their psychological model of reflexion, a facet treatable in thermodynamic terms on both sides, and a shared holographic description.[1] They speculate that intelligent black holes trigger the creation of massive stars, from which further black holes arise after collapse. Graded against the membership criteria of Section 1, this fails both: the Fermi observation plays no role, and the grounding is structural analogy rather than the thermodynamics of computation, so Figure 2 carries it as a grey precursor and the family count is unchanged. We cite it for priority; the sweep found no later work in the lineage building on it.

One grey-literature item postdates even the newest member and is recorded here for completeness. Kriger (2026), a Zenodo preprint dated 31 January 2026 and not peer reviewed, argues the inward turn directly: computational efficiency acts as a universal attractor of civilizational development, advanced civilizations redirect resources from physical expansion to computation, and the silence follows. That framing meets membership criterion (i). What it lacks is the family's physics: the argument runs on efficiency contrasts (transporting mass costs of order mc^2, processing information of order $k_{\mathrm{B}}T$) without naming a substrate, a site, or an observable, and it distinguishes itself from transcension by disclaiming convergence toward physical optima such as black holes; it has no star-cluster content. It is entered as a precursor of the framing rather than a member, and the count, matrix, and scorecard are unchanged.

3.2 Ćirković & Bradbury (2006): the migration hypothesis

Claim. Advanced civilizations are postbiological and migrate to the Galactic outskirts, where low temperatures favour efficient computing; SETI fails because it searches the wrong place (inner Galaxy, habitable zones) for the wrong thing (biological-era beacons) (Ćirković and Bradbury, 2006). *Mechanism.* Two gradients, not one: the same Landauer logic as Section 2.3 applied radially across the Galaxy (ambient temperature falls with galactocentric radius, with the maximum information-processing rate per joule of dissipated free energy, $(I/E)_{\max} \propto \sqrt{R}$ with I the number of elementary operations, rising appreciably beyond $R \gtrsim 10$ kpc on Ćirković and Bradbury (2006)'s own scaling), plus a metallicity leg the authors argue cuts the same direction: the outer disk's lower metallicity ($\nabla Z \approx 0.07$ dex/kpc) is framed as an advantage rather than a penalty, on the reasoning that an advanced civilization synthesizes heavy elements more efficiently than stellar nucleosynthesis, so primordial-composition baryonic matter is desirable feedstock for optimized computing substrate. *Predicted sky.* "City lights" at the rim: low-temperature infrared structures in the outer Galaxy; silence in the traditional SETI bands. *Objections.* The thermodynamic leg is the vulnerable one: the CMB floor of 2.7 K applies everywhere, so on any bound that treats the

[1] For the Soviet and Russian SETI literature more broadly, see the review of Gindilis and Gurvits (2019).

radial gain as flat rather than as the authors' own $\sqrt{R}$ scaling ($\sqrt{25/8.2} \approx 1.75\times$ from the solar circle to a 25 kpc rim), the rim's advantage over the solar neighbourhood is a factor of order unity (1.75×), not the 10^{12+} available from horizon sinks, and the observational bind is real at both ends: structures radiating near the achievable 3–10 K floor peak at 300–1000 μm, outside the WISE bands entirely, so cold-end constraints require far-infrared or submillimetre surveys that have not been performed, while only the warm ($\gtrsim$100 K) end of the predicted population falls within WISE-class sensitivity (the $\hat{\mathrm{G}}$ framework's stated reach is ~150–600 K; colder structures enter only with implausibly large emitting areas), where it has been neither sought as a targeted population nor found serendipitously (Wright et al., 2014a; Griffith et al., 2015). The metallicity leg is the more exposed of the two and has not been engaged here: it assumes in-situ transmutation is cheaper than harvesting already-processed heavy elements from the inner disk, an assumption the authors do not defend against the competing economics of simply relocating refined material outward. *Status.* Untested as stated, but its premise survives intact in every later member; historically the first explicit "wrong place, wrong observable" Fermi solution grounded in computation thermodynamics.

3.3 Smart (2012; 2019): the transcension hypothesis

Claim. The developmental trajectory of complexity, "STEM compression" of space, time, energy, and matter, carries every sufficiently advanced intelligence toward black-hole-density confinement and, ultimately, out of the observable universe; the silence is a developmental constant, not a contingency (Smart, 2012, 2019). *Mechanism.* An evolutionary-developmental analogy: as in ontogeny, the endpoint is convergent and encoded in the dynamics, with black-hole-scale density as the attractor. *Predicted sky.* A named target class, the "galactic transcension zone": an inner ring of each galaxy's habitable zone where civilizations that transcended long ago have collapsed to near-black-hole density, yielding a deficit of life-signature exoplanets in that ring relative to the outer habitable zone; Smart also states a timing claim, that the interval from the cessation of early low-power electromagnetic leakage (radar, radio, television) to reorganization at near-black-hole density runs ~400 yr; transiting-companion signatures are offered as a secondary, long-shot observable. *Objections.* The terminal step (exit from the universe) invokes physics that is unspecified and untestable in principle — the standing criticism that motivated the "macro" variant (Section 3.7); the developmental analogy supplies direction but no selection theory over real objects; and the stated kill conditions are conditional on an inner-habitable-zone exoplanet census at a completeness no funded instrument delivers, so confirmation and exclusivity of the hypothesis is untestable with today's science even though the deficit it predicts is, in principle, a result observation could contradict — Smart's own concession is that "proving the existence and exclusivity of the transcension hypothesis with today's science may be impossible," a narrower claim than unfalsifiability tout court. *Status.* Falsifiable in principle but untested at the completeness its own kill conditions require (Table 3); enormously generative as a research program — nearly every subsequent member cites it as the frame.

3.4 Vidal (2014; 2016): stellivores

Claim. Some observed astrophysical systems, specifically accreting binaries, may already be advanced civilizations feeding on stars: "stellivores." The Fermi question is inverted: they are not absent, they are misclassified (Vidal, 2014, 2016). (Vidal 2011 is an earlier, independent statement of the related black-holes-as-attractor thesis and is discussed with the black-hole-infrastructure thread, Section 3.6, rather than here.) *Mechanism.* Energy logic of accretion (correct, and quantitatively the strongest in the family pre-2023), combined with a cybernetic reading of binary-system regulation. *Predicted sky.* The X-ray-binary and cataclysmic-variable populations contain technological members distinguishable,

if at all, by anomalous regulation statistics. *Objections.* Standard accretion astrophysics explains the cited phenomenology without remainder, so the proposal carries an extreme parsimony burden; Vidal (2016) proposes observational criteria intended to separate stellivore candidates from ordinary accreting binaries, but no statistic with demonstrated discrimination performance against real catalogues has been exhibited (the R accretion-regulation statistic of Swanson 2026b is the same class of statistic, developed for engineered IMBH flows rather than for binaries, and marks the route such a discriminator would take); and the claim's retreat to in-principle indistinguishability flirts with unfalsifiability. *Status.* No proposed test has been carried out; the population-statistics route (regulation anomalies across accreting-binary catalogues) remains open and is the fair test (Section 7).

3.5 Sandberg, Armstrong & Ćirković (2016): aestivation, and the Bennett–Hanson–Riedel critique

Claim. Computation-maximizers harvest resources now but defer computation to the far future, when the CMB has cooled toward the de Sitter floor and each joule buys up to $\sim 10^{30}$ times more erasures; they are dormant now, hence silent (Sandberg et al., 2016). *Mechanism.* The sink hierarchy of Section 2.3, exploited in time rather than in space. *Predicted sky.* Quiet, resource-conserving infrastructure; suppressed stellar-disruption rates in controlled regions; possibly missing baryons curated against waste. *Objections.* The decisive one is thermodynamic: Bennett et al. (2019) showed that erasure can be paid for at any epoch by transferring entropy into existing non-maximal-entropy reservoirs, and that this is not improved by waiting for a colder background, so dormancy confers no advantage. A separate argument, ours, is that free energy not collected promptly is irreversibly lost (stars burn regardless), so the optimal policy is aggressive present-epoch harvesting with storage, not dormancy; stored negentropy can be spent at any later epoch at that epoch's efficiency, so waiting confers no advantage over harvesting now and computing later, and storage itself carries only small maintenance costs. The critique has a second leg that matters for this review: Bennett et al. (2019) also observe that the present universe already contains vast reservoirs far from maximal entropy into which computation-generated entropy can be dumped, so the premise that erasure must be charged at the ambient CMB temperature fails today; that observation amounts to an endorsement of present-epoch cold sinks, which the horizon-sink thread (Sections 2.3 and 3.6) radicalizes to its physical extreme. The authors' premise that erasure cost dominates the budget is also model-dependent: reversible architectures shift the optimum toward computing early. Aestivation survives only in attenuated form (harvest now, archive-grade computation deferred), at which point its observational predictions collapse into those of its neighbours. *Status.* The critique has not been rebutted in print, though Sandberg has responded informally; the hypothesis's lasting contributions are the explicit defence policy, which yields two sharp falsifiable predictions (intervention against entropy-wasting astrophysical processes, and an archival search for gravitationally bound structures in excess of ΛCDM predictions), and the formalization of the efficiency calculus that the whole family now uses.

3.6 Black holes as infrastructure: Inoue & Yokoo, Hsiao et al., Dvali & Osmanov

Claim(s). Black holes are the terminal power plants and computers, argued at three different host-mass scales: collectors around accreting supermassive holes (Inoue and Yokoo, 2011); Dyson-type harvesting of disk luminosity around a stellar-mass hole, with computed detectability (Hsiao et al., 2021); and black holes (including manufactured micro holes) as the most efficient quantum information processors, with a predicted high-energy neutrino signature from their evaporation (Dvali and Osmanov, 2023). Vidal (2011) independently argues the general case, that black holes are attractors for intelligence on a two-dimensional Kardashev–Barrow development metric, without committing to a host-mass scale. A time-dilation variant

has since appeared in Reiss and Feng (2026): civilizations on biologically tolerable orbits near the photon radius of a supermassive hole, maintainable well below the Type II threshold; with a time-dilation factor of 10^4, a single Type II civilization could span the Galaxy within a crew lifetime in its own reference frame, and isotropic signals emitted on such orbits would exhibit a downward frequency drift. *Mechanism.* Kerr accretion efficiency (5.7–30 per cent of rest mass, Thorne-capped at $a_\star = 0.998$; Bardeen et al. 1972; Novikov and Thorne 1973; Thorne 1974), Blandford–Znajek spin extraction (Blandford and Znajek, 1977; Tchekhovskoy et al., 2011), and horizon-saturated information physics. The thread has a technical prehistory older than any family member: Penrose (1969) supplied the rotational-extraction mechanism, Misner et al. (1973, §33.7) formalized the civilization-scale power-station thought experiment around it, and Thorne (1994) and Frolov and Novikov (1998, §12) carried the idea through the popular and textbook literature; Figure 2 records this as a grey precursor lineage feeding the present thread. The thread continued past the family's founding on a second front: Crane and Westmoreland (2009) asked whether starships or power sources could be built on the Hawking radiation of an artificially made black hole, judging the concept to be at the edge of possibility, which extends the harvesting route from rotational extraction to evaporation. *Predicted sky.* Inoue and Yokoo (2011): a deliberate, sharp coherent electromagnetic beam transmitting harvested energy, a voluntary-class signature and the thread's earliest and most direct observable. Hsiao et al. (2021): a hot ($\sim 9.5 \times 10^4$ K) thermal excess superposed on an otherwise standard accretion-disk SED around a stellar-mass hole, detectable in the ultraviolet through mid-infrared at Galactic distances with GALEX/HST-class photometry, discriminated by SED model-fitting plus radial-velocity variability. Dvali and Osmanov (2023): tens-of-TeV neutrinos and photons, in bursts for single-computer events and as a continuous flux for populations of simultaneous computations, detectable against IceCube- and MAGIC-class sensitivity; a genuinely novel observable since natural sources do not readily mimic it (Paper C operationalizes this as a TeV–PeV band with multiplet criteria, an extension beyond the source paper's own tens-of-TeV figure, stated here explicitly). The micro-hole channel inherits a live controversy over whether radiation-collapse manufacturing is blocked by Schwinger pair production (Álvarez-Domínguez et al., 2024; Loeb, 2024). *Objections.* The selection problem here is not the absence of a stated mass scale; each member states one, and the three disagree (supermassive, stellar-mass, manufactured-micro) and none argues for its choice against the other two. A consequence for Table 3: the BH-computing row is graded best-leg across the three proposals (Inoue and Yokoo 2011's galactic-club framing carries C1, Dvali and Osmanov 2023's neutrino channel carries C3, Hephaistos-class Dyson searches bound the collector variant's C5), so the row's 3.5-point total is a composite upper envelope no single published proposal claims on its own; the SMBH orientation of Inoue and Yokoo (2011) in particular ignores the hazard and competition profile of galactic nuclei; detectability claims assume the civilization tolerates conspicuous luminosity, in tension with the efficiency motives that brought it there (though Inoue and Yokoo (2011)'s beamed channel is deliberately conspicuous by design, not an involuntary leak, which is the exception noted below); and the horizon-as-sink accounting inherits the radiator-temperature objection of Section 2.3, adjudicated there as ⊙-open: the carrier-delivery reply applies to this thread as much as to the MTH, and the re-interception step is equally unpriced. *Status.* The Dvali and Osmanov (2023) neutrino channel is the family's most testable near-term prediction and is now the subject of a proposed dedicated monitoring program at a named target (Swanson, 2026d); the high-energy SETI frame is surveyed by Lacki and DiKerby (2025), and the Dyson Minds workshop (Curtis et al., 2026) has since developed the observational agenda for the supermassive-host case, recommending anomaly-detection mining of WISE, JWST, and EHT archives. The compact-object case now also appears in the community review of technosignature searches: Vidal et al. (2026) close their stellar section with compact objects as attractive sites for advanced civilizations, black holes among the attractors for intelligence, and list as a signature a black hole whose rotational rate is slowing down anomalously, i.e. a sudden change in angular

momentum (their §5.5), the spin-residue channel graded in Section 7. The distinction between voluntary and involuntary technosignatures (Haliki, 2025) is useful here, with one exception stated plainly: Inoue and Yokoo (2011)'s galactic-club framing predicts a deliberate beamed signal, not a by-product, so the thread is not uniformly involuntary; for the efficiency-motivated members (Hsiao's collectors, Dvali's computers, and the MTH), an efficiency-driven occupant emits no beacons, so those observables are involuntary by-products, the class the residue catalogues of Sections 3.7 and 6 enumerate.

3.7 The Macro Transcension Hypothesis (2026)

Claim. The inward trajectory halts at a macroscopic, observable destination: rapidly spinning IMBHs embedded in dense, old, low-luminosity stellar systems, selected by a four-axis gradient (energy per unit fuel, entropy disposal, storage density, clock-rate control); migration and operation are electromagnetically quiet; Omega Centauri is the highest-ranked accessible target (Paper A). *Mechanism.* The full Section 2 stack, plus a selection theory over real objects: intermediate mass (storage and erasure advantages saturate at $10^{4–5}\ M_\odot$ while hazards grow), high spin (extractable energy), dense old clusters (fuel, stability, uncontested claim). *Predicted sky.* Silence plus residues: anomalously high spin on gas-starved IMBHs; suppressed accretion relative to ambient supply; secular core depletion; possibly Dvali–Osmanov neutrino bursts. Six instrument-matched tests, of which T3 and T5 are the decisive kill tests and T1, T2 and T6 are constraint channels that no accumulation of electromagnetic non-detections can turn into a refutation (Paper A's falsification-test schedule, Table 7 as of v2.9; operationalized in Paper C). Where this review cites ω Cen's own cluster parameters, see Paper F App. A for the shared-parameter table across the set. *Objections* (catalogued at full strength, given the disclosure of Section 1): it inherits the optimization-pressure premise unargued (Section 6); the staged architecture presupposes goal stability over 10^8 yr; the flagship target's IMBH is itself contested ($\lesssim 6 \times 10^3\ M_\odot$ timing bound versus $\geq 8.2 \times 10^3\ M_\odot$ kinematic bound; Bañares-Hernández et al. 2025; Häberle et al. 2024; the latest joint MeerKAT–Parkes pulsar timing, spanning 2021–2025, is insensitive to an IMBH of 10^3–$10^4\ M_\odot$ and constrains the central mass only to $< 10^5\ M_\odot$ at 90% confidence, Colom i Bernadich et al. 2026, leaving the tension unresolved); the class-level retreat (47 Tuc, M54, extragalactic nuclei) risks unfalsifiability-by-relocation, partially bound by the stated class prediction (no Galactic cluster IMBH with $a_\star \gtrsim 0.9$ falsifies it for the Milky Way); its erasure-cost advantage rests on the contested horizon-as-sink accounting of Section 2.3, adjudicated there as ⊙-open: the carrier-delivery reply is Paper A's own, so the unpriced re-interception step binds the MTH before any other member; and the spin residue, its sharpest discriminator, waits on LISA (Babak et al., 2017; Colpi et al., 2024). *Status.* Untested; distinguishable from its relatives chiefly by having purchased falsifiability with named targets, thresholds, and timelines — the property the scorecard (Section 4) is designed to price.

3.8 The family at a glance

Table 1 condenses the six member treatments of this section into a single view and appends each member's scorecard row, so that the qualitative comparison above and the quantitative grading of Section 4.2 can be read side by side. The cells are condensed from the subsections, not new material; the symbols in the last column are those of Table 3.

Table 1: The inward family at a glance, condensed from Sections 3 member treatments. Last column: the falsifiability scorecard row in C1–C6 order (• met, ⊙ partial, ∘ not met; † a leg supplied by this review or by critics rather than by the hypothesis's proponents; Table 3).

Hypothesis	Core assumption	Energy logic	Predicted appearance	Key objection	Status	Scorecard row
Migration (Ćirković and Bradbury, 2006)	postbiological optimizers migrate down Galactic temperature gradients	ambient-T Landauer gain toward the rim ($\sqrt{R}$ scaling) plus a metallicity feedstock leg	low-temperature infrared "city lights" at the outer Galaxy; silence in conventional bands	radial leg is CMB-floored to $O(1)$ on a flat reading; transmutation-cheaper-than-harvesting premise undefended	untested as stated	⊙∘⊙∘∘⊙
Transcension (Smart, 2012, 2019)	convergent developmental trajectory (STEM compression) to black-hole density and exit	density conversion toward Lloyd-limit serial computing	transcension-zone exoplanet deficit; ~400 yr from radio silence to black-hole-density reorganization; near-total silence	terminal step untestable in principle; stated kill conditions outrun any funded census completeness	untested at required completeness	⊙∘∘⊙∘∘
Stellivores (Vidal, 2014, 2016)	some accreting binaries are civilizations feeding on their stars	direct stellar accretion under binary regulation	anomalous regulation statistics within X-ray-binary and cataclysmic-variable populations	ordinary accretion astrophysics explains the phenomenology; no statistic with demonstrated discrimination	open; population test feasible	•∘⊙† ∘∘∘
Aestivation (Sandberg et al., 2016)	erasure cost dominates the budget; computation deferred to the cold far future	waiting for the de Sitter floor, up to ~ $10^{30}\times$ erasures per joule	curated quiet regions; suppressed entropy-wasting processes; possibly missing bound structures	waiting does not lower erasure cost (Bennett et al., 2019); reversible architectures shift the optimum	attenuated by the critique	⊙⊙⊙•∘⊙
BH computing (Inoue and Yokoo, 2011; Hsiao et al., 2021; Dvali and Osmanov, 2023)	black holes are terminal power plants and processors across stated host-mass scales	Kerr/Blandford–Znajek extraction; horizon-saturated storage and processing	coherent beamed transmission (supermassive); hot SED excess (stellar-mass); tens-of-TeV neutrino bursts (micro)	three host-mass choices with no comparative selection theory; radiator-temperature sink access open on re-interception	neutrino channel newly testable	⊙⊙•⊙⊙⊙
MTH (Paper A)	a subset of civilizations maximizes computation and relocates along the gradient	full thermodynamic stack plus a selection theory over real IMBH hosts	electromagnetic silence plus spin, accretion-suppression, and depletion residues; possible neutrino bursts	optimization premise unargued; 10^8-yr goal stability; flagship IMBH contested; re-interception unpriced	six pre-registered tests pending (T3, T5 decisive)	••⊙•⊙⊙

Table 2: The inward family compared. "Energy logic" names the dominant free-energy mechanism; "appearance" is the predicted present-epoch sky.

Hypothesis	Core assumption	Energy logic	Predicted appearance	Standing objection	Status
Migration (Ćirković and Bradbury, 2006)	postbiological evolution; T-sensitive computing	ambient-T Landauer gain at Galactic rim ($\sqrt{R}$ scaling) + metallicity-gradient feedstock leg	IR "city lights" in outer Galaxy; band silence	thermodynamic leg is CMB-floored at $O(1)$ on a flat reading; metallicity leg's transmutation-cheaper-than-harvesting assumption undefended	untested as stated
Transcension (Smart, 2012)	convergent STEM compression	density → Lloyd-limit computing	transcension-zone exoplanet deficit; near-total silence; transit anomalies	terminal step untestable; stated kill conditions outrun any funded census	untested at required completeness
Stellivores (Vidal, 2014, 2016)	some binaries are civilizations	direct stellar accretion	misclassified X-ray binaries	no statistic with demonstrated discrimination performance (Vidal, 2016)	open; population test feasible
Aestivation (Sandberg et al., 2016)	erasure dominates budget; patience	wait for T_{dS}: $\sim 10^{30}\times$ gain	curated quiet; process suppression	waiting does not lower erasure cost (Bennett et al., 2019)	attenuated by critique
BH computing (Inoue and Yokoo, 2011; Hsiao et al., 2021; Dvali and Osmanov, 2023)	holes are optimal processors	Kerr/BZ extraction; horizon information	anomalous SEDs; ν bursts	no selection theory over holes; radiator-limited sink access (§2.3)	ν channel newly testable
MTH (Paper A)	subset of civilizations maximizes computation	full stack + selection over IMBHs	silence + spin/depletion residues; ν bursts	optimization premise; goal stability; contested target IMBH; radiator objection ⊙-open: carrier delivery answered, re-interception unpriced (§2.3)	six pre-registered tests pending (T3, T5 decisive)

4 Comparative analysis

4.1 The matrix

Table 2 aligns the family on five axes. Reading across rows reveals the internal disagreements that casual conflation hides: aestivation and transcension disagree about *when* (future versus now); transcension and the MTH disagree about *where* (fabricated density versus existing holes); stellivores and everyone else disagree about *visibility* (already visible versus invisible); and the black-hole-computing thread disagrees with the MTH about *which holes* (supermassive and manufactured-micro versus intermediate).

One asymmetry in the "core assumption" column requires disclosure. The MTH row states its premise in weakened subset-form ("a subset of civilizations maximizes computation") while its competitors are stated in the universal forms their authors used. The subset-form is what Paper A asserts, so the row is accurate, but the weakening buys explanatory cover that the universal forms do not enjoy, and it must be paid for: a subset-form premise faces the incomplete-compliance objection at full strength: partial compliance thins the expected loud population without collapsing the explanation outright, but non-adopters remain to be explained, and full dissolution of the silence puzzle requires either high compliance across the population or supplementary rarity from conventional Drake factors (P4, Section 6). Readers comparing rows should either read the competitors charitably in whatever subset-form their authors would accept, or charge the MTH the P4 penalty; the matrix should not be read as granting the newest member a weaker burden.

4.2 The falsifiability scorecard

Table 3 grades each member against six criteria. Five were fixed before grading in the original version of this review; the sixth, C6, was added in revision at a referee's prompting, and we disclose that provenance here rather than absorb it silently, because the scorecard's claimed virtue is criteria stated in advance and a quietly appended criterion would falsify that claim. The criteria: **C1** – names a target or target class that observations can interrogate; **C2** – states quantitative observables with thresholds; **C3** – at least one stated test is executable with existing or funded instruments within ∼15 years, where "stated" means stated by the hypothesis's proponents; where the executable test was supplied by this review or by critics rather than by the proponents (the stellivore population statistic, the attenuated-aestivation suppression audit), the grade records that provenance; a cell carrying such a leg is marked †; **C4** – specifies kill conditions in advance (results the proponents agree would refute it); **C5** – has survived at least one executed test that could have refuted it, with partial credit (⊙) where executed adjacent tests constrain the hypothesis without having been designed for it; **C6** – resistance to unfalsifiability-by-relocation: when a stated test fails, the hypothesis either accepts refutation or retreats along a pre-bounded fallback that a finite series of nulls exhausts, rather than relocating indefinitely in target, strength, or observable. Symbols: • met, ⊙ partially met, ∘ not met. Where a criterion is tested by a multi-part test, the cell grades on the best executable part unless the parts are jointly necessary to the prediction, in which case the weakest part sets the grade; demonstrated discrimination performance against real catalogues is recorded as a reason a stated test fails executability under C3 rather than as a criterion of its own. Three caveats apply. First, C5 is the column that separates science-in-progress from science-performed: no member of the family scores • on C5, because no pre-registered inward-family test has yet been executed to completion; the FAST globular-cluster survey (Huang et al., 2026b) and the proposed ω Cen campaign (Swanson, 2026d) are the first entries in that ledger. The partial credits record real constraint from tests executed for other reasons: the collector variants of black-hole computing are bounded by Hephaistos-class Dyson searches (Suazo et al., 2024; Ren et al., 2025), and the accretion-limit measurements at ω Cen (Haggard et al., 2013; Chen et al., 2026; Mahida et al., 2026) already execute part of the MTH's suppressed-accretion test series; the migration hypothesis carries no such credit, since its warm-end $\hat{\mathrm{G}}$-survey band has been neither sought as a targeted population nor found serendipitously (Section 3). One conditioning clause on that last credit: it attaches to the limit-tightening restatement of Paper A's test T1, since the originally registered trigger threshold sat several orders of magnitude above achievable sensitivity and is restated in the current revision of Paper A; the ⊙ is contingent on that restatement, marked * in Table 3 and tested against the as-registered reading (∘) under coding K4 below. Second, given the author's conflict of interest, the MTH row's grades are tied to specific Paper A artifacts rather than to judgement: C1 rests on the named flagship target and ranked fallback class (ω Cen; 47 Tuc, M54); C2 on the thresholds Paper A attaches to its LISA tests in its falsification-test schedule, a spin measurement $a_\star < 0.1$ at $> 3\sigma$ (T5) and an EMRI point mass below the kinematic floor, $M < 8.2 \times 10^3\, M_\odot$ (T4); the near-extremal spin $a_\star \gtrsim 0.9$ is not credited, because Paper A expects it on the null hypothesis as well as on the MTH. The T5 credit is branch-conditional: on a MAD-mode spin-up branch (Paper E, Section 3.4) the equilibrium spin is $a_\star \approx 0.035$–0.07, at or below Paper E's natural-capture null and below the T5 trigger $a_\star < 0.1$, so the T5 leg of the C2 credit holds only on the thin-disc spin-up branch; the T4 threshold does not depend on the branch. The C2 credit sits alongside the quantitative ω Cen mass-tension table (Paper A's mass-constraint table, Table 6 as of v2.9) as a secondary, conventional-astrophysics observable rather than the credited threshold itself (no single Paper A table currently tabulates both mass and spin jointly); C3 on the LISA legs of the instrument-matched test schedule (T3, T5; Paper A's falsification-test schedule, Table 7 as of v2.9); and C4 on the two decisive tests of that same table. So that these grades are auditable without leaving this paper, the kill conditions credited are T3, a LISA signature of an extended remnant population or multiple low-mass inspirals rather

Table 3: Falsifiability scorecard. Criteria C1–C6 defined in Section 4.2; C6 was added in revision at referee prompting, as disclosed there. • met, ⊙ partial, ∘ not met. A † marks a criterion satisfied partly by a test supplied by this review or by critics rather than by the hypothesis's proponents (Section 4.2). A * marks a credit contingent on a post-registration threshold restatement: MTH's C5 is ⊙ only under Paper A's limit-tightened T1; as originally registered the credit is ∘, giving MTH a total of 4.0 rather than 4.5 (still first; Section 4.2). The MTH row is graded against Paper A v2.9: C2 rests on T4 and T5, C4 on T3 and T5, and C3 is partial because only the LISA legs count (Section 4.2). The scorecard grades testability, not truth.

Hypothesis	C1 target	C2 thresholds	C3 near-term	C4 kill criteria	C5 tested	C6 relocation
Migration (2006)	⊙	∘	⊙	∘	∘	⊙
Transcension (2012)	⊙	∘	∘	⊙	∘	∘
Stellivores (2014–16)	•	∘	⊙†	∘	∘	∘
Aestivation (2016)	⊙	⊙	⊙	•	∘	⊙
BH computing (2011–23)	⊙	⊙	•	⊙	⊙	⊙
MTH (2026)	•	•	⊙	•	⊙*	⊙

than a single point mass, which removes H_2 entirely, and T5, a LISA spin measurement $a_\star < 0.1$ at $> 3\sigma$, which removes the exploitation model. Paper A (v2.9) classes T1, T2 and T6 as constraint channels, not kills, and T4 as a demotion, not a kill, and states that no accumulation of electromagnetic non-detections, however deep, can falsify the MTH; none of the four is credited under C4. C3 is graded on the same two tests and is partial: LISA is funded and its mission window falls partly inside the ∼15-year horizon, but the per-cluster event rate is uncertain by orders of magnitude (Section 7), so execution is conditional on an inspiral occurring. The JWST, Chandra, MeerKAT and KM3NeT legs (T1, T2, T6) are executable today, but C3 is read here as requiring a stated test that can adjudicate H_0 against H_2, and Paper A states that T1, T2 and T6 cannot (those hypotheses predict identical electromagnetic appearances), so these legs carry no C3 credit. A reader who credits them restores the full grade, a reading priced in the codings paragraph below, which also records the re-grade against v2.9. Third, the MTH's strong showing on C1, C2 and C4 was purchased after the criteria of falsifiability had been articulated by its predecessors' critics; later entrants always grade better on form, and the scorecard measures testability, not probability of truth.

Four scorecard judgements require defence. *Migration's* ⊙ on C3 is proponent-stated, not review-supplied: Ćirković and Bradbury (2006) themselves recommend infrared surveys for Dyson shells and Matrioshka-class structures at the Galactic rim, and the census in Table 4 operationalizes that recommendation directly, splitting it into the warm ($\gtrsim$100 K) leg WISE already covers at area-limited depth and the cold (3–10 K) leg that no flown or funded far-infrared/submillimetre survey reaches. The half-credit reflects the warm leg's near-term executability against a stated zero-excess-candidate threshold, offset against the cold leg's lack of any funded instrument; it is not a claim of proponent-stated completeness across the whole predicted population. *Aestivation's* • on C4 reflects Sandberg et al. (2016)'s explicit and admirable statement of refuting observations (uncontrolled entropy-wasting processes in regions a defender should police); its ⊙ on C3 splits two stated tests, both Sandberg et al. (2016)'s own: the process-suppression audit, which no instrument can execute at the required scale on any near-term horizon (∘ on that leg), and the proposed archival test, a search for gravitationally bound structures more massive than ΛCDM predicts, on the reasoning that halo-scale curation would be rational and visible and its absence bounds energy-concerned aestivators (• on that leg, executable against existing survey catalogues); the two average to ⊙. This is a weakest-leg-average rule applied within a single proposal's two stated tests, a different rule from BH computing's best-leg envelope across three independent proposals (above); the two rules are not interchangeable, and the choice is disclosed rather than harmonized because the two rows differ in kind, one proponent with two legs against three proponents with one leg each, but a reader applying best-leg uniformly would move aestivation's C3 to •, tying it with BH computing at 3.5 for clear

second place instead of trailing at 3.0. The "archival/feasible now" entry in Table 4 operationalizes the mass-function leg; the process-suppression leg and the attenuated form's collapsed predictions (Section 3.5) remain the harder, untested core. *Transcension's* ⊙ on C1 and C4 credits the named transcension-zone target class and Smart's own stated kill condition (a persistent inner-habitable-zone life-signature population); it is not raised further because the completeness that condition requires sits beyond any funded census, and because the terminal step, exit from the universe, remains unobservable in principle. This is not a dismissal of the hypothesis's generativity (half the family descends from it) but the direct application of the criteria to a proposal whose falsifiable core (the exoplanet deficit) is real and stated, while its endpoint is not; Smart (2019) adds developmental argument, not additional observables. *BH computing's* • on C3 is carried entirely by the Dvali and Osmanov (2023) neutrino channel, which predicted a high-energy neutrino signature from a source class that current telescopes can monitor (The KM3NeT Collaboration, 2025; Aartsen et al., 2020); the specific energy band and multiplet criteria are as operationalized in Paper C. Finally, the MTH and aestivation now both grade ⊙ on C3. The MTH's credited tests are the LISA legs (Table 4), which are funded but depend on an in-band inspiral whose per-cluster event rate is uncertain by orders of magnitude (§7); its T1, T2 and T6 legs are executable today but are constraint channels in Paper A's v2.9 classification and carry no C3 credit. Aestivation's half-credit comes from the mass-function leg alone, and the process-suppression leg that would close the gap has no equivalent funded instrument on any announced schedule.

C6, the revision-stage criterion, requires its own defence, since it grades the axis on which this review's prose is hardest on its own member. The MTH's class-level retreat (47 Tuc, M54, extragalactic nuclei; Section 3.7) is bounded within the Galaxy by a stated class prediction that a finite series of spin measurements exhausts, but the extragalactic fallback remains unbounded: ⊙. Aestivation's response to the Bennett et al. (2019) critique was retreat to the attenuated form, which is relocation-in-strength; its full-strength kill condition remains on the table, so it also grades ⊙. Stellivores' appeal to in-principle indistinguishability is relocation-in-observable, with no bounded fallback stated: ∘. Transcension's endpoint is defined as unobservable, so there is nothing to relocate from and nothing to resist with: ∘. The migration hypothesis makes a population-level claim over the whole outer Galaxy and can retreat in temperature and depth though not in place: ⊙. BH computing grades ⊙: the Dvali and Osmanov (2023) channel closes at a stated strength, but the thread's missing selection theory over holes leaves target relocation open. The column's outcome is uncomfortable for the reviewer in a useful way: the newest member shares the column's top achieved grade with three of its relatives, ahead of two, and no member meets the criterion outright.

The half-grade handling is worth testing rather than asserting. Re-grading Table 3 under four alternative codings, recorded in advance in the accompanying script (`figs/fB_calc7_v1.py`): counting a partial as unmet everywhere (K1); counting it as met everywhere (K2); replacing the one documented two-leg average (aestivation on C3) with its weaker leg (K3); and withdrawing MTH's contingent C5 credit, the half-grade tied to the limit-tightening restatement of Paper A's T1 threshold rather than to the threshold as originally registered (K4). An earlier version of K4 withdrew the migration hypothesis's own C5 partial; that partial is already ∘ in the printed table above (its warm-end search was neither sought nor found, Section 4.2), so the coding was a no-op and K4 now targets the one contingent half-grade the table still carries. Baseline (Table 3 as printed): MTH 4.5 (1st), BH computing 3.5 (2nd), aestivation 3.0 (3rd), migration and stellivores tied at 1.5 (4th–5th), transcension 1.0 (6th). The v2.9 re-grade of the MTH row lowers its total from 5.0 to 4.5 and its margin over BH computing from 1.5 to 1.0 points; the MTH stays first. It ranks first singly under K1 (3.0, against 1.0 for each of the next three), K3 (unchanged at 4.5), and K4 (4.0, a 0.5-point margin over BH computing), and ties BH computing under K2 (6.0 each, the reading that converts every partial to a full grade), as before the re-grade. If the T1, T2 and T6 legs were credited

under C3 despite the classification above, the baseline would return to 5.0 and K1, K3 and K4 to 4.0, 5.0 and 4.5. The mid-table is less stable: K1 flips the migration/stellivore order (stellivore rises to a three-way tie with aestivation and BH computing at 1.0, while migration falls to last, tied with transcension at 0.0), and K2 ties stellivore with transcension at 2.0. On C6 specifically, no coding separates the column's four ⊙ grades or lifts its two ∘ entries, so the closing judgement above does not depend on how partial credit is counted.

5 The sociological competitors

Three well-known solutions predict the same silent sky from non-thermodynamic premises, and the family is best understood against them. A fourth class, the expansion-slowing models, sits between the sociological solutions and the inward family and is treated last.

Zoo (Ball, 1973) and its hegemony refinement (Forgan, 2017). Silence as policy: we are observed but quarantined. Forgan's contribution is structural — a zoo requires either a single dominant culture or stable inter-civilization consensus ("Galactic Club"), and fragmented "cliques" should leak defectors. The premise is sociological uniformity over megayear timescales, the premise the inward family replaces with physics.

Dark forest (Brin, 1983; Yu, 2015). Silence as survival strategy under first-strike game theory; the name comes from Liu Cixin's 2008 novel, and the academic formulation is due to Yu (2015), with Brin (1983) as the earlier "deadly probes" antecedent. Its physical premises are at least defensible (relativistic kill vehicles, asymmetric detection), though the kill-vehicle and targeting literature is disputed on interstellar targeting accuracy, light-lag, and deterrence grounds, and its deadly-probes mechanism has received technical treatment: Forgan (2019a) models mutated "predator" probes suppressing a visible probe population with Lotka–Volterra dynamics and finds that stable equilibria typically retain large prey populations, undercutting the mechanism as a full silence-explanation. The equilibrium is also fragile: the strategy must be adopted by every civilization without exception and without deviation forever, since a single loud defector (or a single expansionist willing to absorb risk) breaks the explanatory account (Brin, 1983; Forgan, 2017). Like the zoo, it predicts pure silence, with no residue and no scheduled adjudication.

Sustainability (Haqq-Misra and Baum, 2009). Exponential expansion is self-terminating; civilizations that persist are those that grow slowly or not at all. This is the mildest competitor, and partially complementary: it removes the pressure to expand without supplying a positive account of where mature capability is directed.

Expansion-slowing models (Landis, 1998; Carroll-Nellenback et al., 2019). A distinct class predicts partial silence from the dynamics of expansion itself rather than from psychology or policy. Landis (1998) models colonization as a percolation process: with finite settlement probabilities and range limits, expansion fronts can stall and leave permanent unvisited voids, one of which could contain the Sun. Carroll-Nellenback et al. (2019) couple settlement fronts to stellar motions and planetary habitability windows and find steady states in which the Galaxy is substantially but incompletely settled, with resettlement lag producing local voids (the "Aurora effect"). These are the inward family's closest rivals among silence-explanations, closer than the zoo or the dark forest, because they are grounded in physics and dynamics rather than in universally adopted strategy, and they are falsifiable in population statistics (void geometry, settlement-front morphology, the age distribution of settled systems). They differ from the inward family in what they explain: they account for local emptiness while leaving expansion as the governing drive, whereas the inward family removes the drive itself. The two are compatible, and a mixed population would show both signatures. They are excluded from the scorecard (Section 4.2) on the same membership ground that excludes Ćirković (2008) from the family itself (criterion (ii), Section 1):

both supply a dynamical or percolation mechanism, not the thermodynamic-optimization mechanism the scorecard's criteria are built to grade. The comparative ranking of Table 3 is accordingly conditional on that boundary, not a claim that the two classes fare worse against the same criteria.

The comparison clarifies what the inward family is actually claiming. All these proposals, sociological and thermodynamic alike, break the expansionist premise. The differences are two. *Robustness*: dark-forest and zoo equilibria are fragile to a single defector among all civilizations across all time, whereas the inward account requires no coordination at all — each civilization independently follows its own resource gradient, and partial compliance thins the expected loud population without collapsing the explanation outright, though non-adopters remain to be explained and full dissolution requires either high compliance across the population or supplementary rarity from conventional Drake factors (P4, Section 6; the thinning made quantitative in the grabby-aliens frame, Hanson et al. 2021). *Residue*: fear and policy predict nothing observable, by design; thermodynamic optimization is a physical process operating on physical objects, and physical processes leave traces: spin histories, depletion statistics, suppressed accretion, burst channels. On both counts the expansion-slowing models score with the inward family rather than with the sociological solutions: they need no coordination and they predict population-level statistics that surveys can test. Among the silence-explanations, these two classes form the falsifiable-in-principle wing, and grading its members' falsifiability in practice (Table 3) is therefore where a review can add value.

6 Open problems

We state the family's unsolved problems as research questions rather than objections, since several are tractable.

P1: The optimization premise. Every member assumes some civilizations converge on maximizing computation (or a correlate). The premise is defended by universality arguments (computation is the currency into which most terminal goals are convertible; Smart 2012; Sandberg et al. 2016; Bostrom 2003) but it has never been given a selection-theoretic foundation: under what competitive or evolutionary dynamics do computation-maximizers come to exist, persist, and dominate the behaviour of their lineage? The grabby-aliens machinery (Hanson et al., 2021; Olson, 2015) could be adapted to model mixed populations (maximizers, expansionists, satisficers) and derive the observable consequences of each mixture; the companion economics paper supplies a capture term for that machinery but defers the full integration (Swanson, 2026a). The physical side of the premise has begun to accumulate its own literature. Wright (2023) applies the Landsberg formalism to Dyson spheres as work extractors and computational engines, finding little to no advantage in nesting shells and an optimum in small, hot spheres, which prices the radiator constraint the Bradbury lineage discovered. Garrett (2026) work the cold end without horizons: computation-dominated civilizations may dissipate at temperatures far below the 100–600 K window of existing waste-heat searches, approaching the CMB floor at large circumstellar radii, with the required radiating area growing as T^{-4}; the predicted "Slysh halo," a partial Dyson swarm radiating grey, line-free thermal emission in the far infrared and submillimetre, is constrainable from archival surveys at the 10^{20} W level, a cold-sink route that needs no horizon.

P2: Goal stability over 10^8 years. Inward strategies require coherent purpose across timescales exceeding mammalian evolutionary history, flagged as the largest unargued premise even by the family's own proponents (Paper A). The mitigation argument (front-loaded payoffs leave permanent residues even if the lineage fragments mid-program) rescues the observational program but not the hypothesis's

interior logic. The family's deepest dependency is therefore a theory of institution or value stability under self-modification, a problem that lies outside astronomy.

P3: Migration economics. Bostrom (2003)'s opportunity-cost argument prices delay; Bennett et al. (2019) price dormancy; nobody has priced relocation. The trade is concrete: a migrating lineage abandons its accumulated local infrastructure for transit times of $10^{4–6}$ yr in exchange for a destination whose advantages (Section 2) are enormous but deferred. Under what discount rates, risk models, and replication strategies does migration dominate local densification (Matrioshka-style; Bradbury 1999)? This is a well-posed optimization problem, and it has since received a first quantitative treatment in a companion paper (Swanson, 2026a), which derives closed-form crossover thresholds among staying, migrating, and seed-dispatch strategies. We retain the problem in this list because the treatment's inputs (discount and hazard rates over agents never observed) remain open, and because its population-level corollaries feed directly into P4.

P4: Incomplete compliance. If migration is optional, the loud expansionists we do not see still need explaining. The defensible position (argued in Paper A): partial compliance thins the expected loud population without collapsing the explanation outright, since each civilization independently follows its own resource gradient, but non-adopters remain to be explained, and full dissolution requires either high compliance across the population or supplementary rarity from conventional Drake factors, not thinning alone. Quantifying "how much thinning suffices" against the Hanson et al. (2021) selection effect is open problem P1's observational face.

P5: Degeneracy of silence. H_0-type explanations (gas starvation, rarity) and inward-family explanations predict identical electromagnetic appearances; only residues discriminate. The family therefore lives or dies on the residue catalogue (spin, depletion, bursts, regulation statistics), and enlarging that catalogue with new discriminators is the highest-leverage theoretical work available. A companion paper takes a first step by forward-modelling two such discriminators for engineered IMBH systems, a waste-heat floor and an accretion-regulation statistic, within a hierarchical Bayesian adjudication framework (Swanson, 2026b). The contested micro-black-hole manufacturing channel (Álvarez-Domínguez et al., 2024; Loeb, 2024) is a case in point: it would be a sharp discriminator if the manufacturing route is physical, and it drops out of the catalogue if Schwinger pair production blocks it.

7 The observational program

The family is no longer untestable in practice; Table 4 maps each member to its executable tests. Four clusters of activity dominate.

(i) Globular clusters as the family's natural laboratory. Old, dense, dynamically relaxed, electromagnetically quiet, and, for the IMBH-hosting subset, equipped with the objects the gradient selects. The niche was identified by Di Stefano and Ray (2016), who argued that globular clusters are "sweet spots" for long-lived civilizations on habitability and stellar-density grounds and proposed them as SETI targets; the first dedicated globular-cluster survey (Huang et al., 2026b) operationalized it a decade later. The habitability argument is independent of and complementary to the thermodynamic-gradient argument of Section 2: the two select the same targets for different reasons. Gaia-based dynamical-extremeness metrics now exist for ranking clusters as technosignature targets: the phase-space crystallization index of Huang et al. (2026a), computed across 79 Galactic globular clusters, places ω Cen and 47 Tucanae among the small high-structure tail of the population. Two properties of that result matter for this review. It is independent

Table 4: Instrument-matched tests of the inward family, 2026–2040. Status: A = archival/feasible now; P = proposed; F = free by-product of funded missions. A parenthetical on a status entry scopes it to the leg of the test that carries the grade.

Hypothesis	Test	Instrument(s)	Status	Refutes / constrains
Migration (rim)	targeted IR census of outer-Galaxy low-T structures	WISE/JWST archival; Roman	P (warm leg, $\gtrsim$100 K, A/area-limited on WISE; cold leg, 3–10 K / 300–1000 μm, P, no far-IR/submm survey exists)	rim "city lights" population, at a stated pass/fail threshold of zero excess candidates above the WISE-class sensitivity floor across the surveyed outer-Galaxy footprint; WISE is all-sky at shallow depth while the JWST archive covers only pointed fields, so the deep leg of the census is not archival over the outer-Galaxy area; faces the confusion-limited systematics of the Hephaistos episode (§1.1)
Stellivores	regulation-anomaly statistics across accreting-binary catalogues	X-ray archives (Chandra, XMM, eROSITA)	A	"feeding" vs "fed" discrimination
Aestivation (attenuated)	process-suppression statistics in old quiescent systems; separately, a census for gravitationally bound structures in excess of ΛCDM predictions (Sandberg et al., 2016)	survey archives; LSST	A (mass-function leg)	curated-region prediction; process-suppression leg untested near-term, mass-function leg archival now — full-strength hypothesis graded $\odot$ on C3 (Table 3), and the attenuated form's predictions collapse into its neighbours' (§3.5)
BH computing (ν)	burst-mode TeV–PeV multiplets from candidate sites	KM3NeT/ARCA, IceCube	P	Dvali and Osmanov (2023) channel at stated strength
MTH	six tests at ω Cen (Paper A v2.9): accretion and waste-heat limits (T1, T2), LISA remnant-population, mass and spin (T3–T5), neutrino bursts (T6)	JWST, Chandra, MeerKAT; LISA; KM3NeT/ARCA	P/F	kills T3, T5; T4 demotion; T1, T2, T6 constraint channels (Paper A)
Family-wide	waste-heat nulls at increasing depth	$\hat{\mathrm{G}}$ successors; Gaia Dyson candidates	A	expansionist premise (control)
Family-wide	GC technosignature surveys	FAST (Huang et al., 2026b); MeerKAT	A/P	first dedicated old-population SETI

evidence in the residue channel the scorecard prices, model-agnostic, kinematic, and authored outside this paper set; and it is simultaneously a null, since the same analysis finds every recovered substructure consistent with known dynamical processes, excludes missing cold shells above the few-per-cent level by Monte Carlo injection, and frames the index as a search-prioritization tool rather than as evidence of engineering, a framing this review adopts unchanged. The mundane origin of ω Cen's phase-space complexity has direct spectroscopic support in its traced tidal tails and accretion debris (Kuzma et al., 2026). A practical corollary follows from geography: the survey that operationalized the target class (Huang et al., 2026b) cannot reach ω Cen from FAST's latitude, so the metric's top-ranked cluster has never been searched at narrowband depth, and completing that program from the southern hemisphere (MeerKAT, SKA-Mid) is an open observational slot that the campaign of Paper C claims. The ω Cen multi-messenger campaign (Paper C) is the first program designed end-to-end around inward-family falsification, with the conventional astrophysics (the IMBH mass tension; Häberle et al. 2024; Bañares-Hernández et al. 2025) guaranteeing scientific value under every outcome. That tension remains live: the latest joint MeerKAT and Parkes timing of the cluster's millisecond pulsars (2021–2025) is insensitive to an IMBH in the contested 10^3–$10^4\ M_\odot$ range and bounds the central mass only from above, at $< 10^5\ M_\odot$ at 90% confidence (Colom i Bernadich et al., 2026), so the timing–kinematics discrepancy awaits deeper data. The cluster's compact-object inventory has meanwhile begun to resolve directly: 23 years of HST and JWST astrometry have yielded the first stellar-mass black hole detected astrometrically in any globular cluster, the $4.46\ M_\odot$ oMEGACat BH-2 in a 94-yr binary (Whitaker et al., 2026), a single low-mass detection rather than a representative member of the $\sim 10^4$-strong, $31\ M_\odot$-mean remnant population the cluster's extended dark mass implies via the pulsar-timing estimate (Bañares-Hernández et al., 2025; Swanson, 2026b).

(ii) The spin channel. Among all residues, black-hole spin is unique: permanent, quantitative, and measurable to $\sim 10^{-3}$ precision by LISA for favourable systems (Babak et al., 2017; Amaro-Seoane, 2018; Colpi et al., 2024). The precision figure applies to detected EMRIs, and the channel therefore depends on event rate: a compact-object inspiral must occur in-band during the mission, and the per-cluster rate for a $\sim 10^4\ M_\odot$ hole is uncertain by orders of magnitude, which is the spin channel's weak link and part of the P5 accounting. Under Paper A's coherent-growth prior a near-extremal spin is expected on the null as well (Section 4.2), so it discriminates only in conjunction with electromagnetic silence; a low spin ($a_\star < 0.1$, Paper A's T5) retires the exploitation models on the thin-disc spin-up branch. No other proposed technosignature combines this permanence with a funded instrument.

(iii) The neutrino channel. The Dvali and Osmanov (2023) burst phenomenology is testable now at named targets with operating instruments (The KM3NeT Collaboration, 2025; Albert et al., 2020); the pre-registered multiplet criteria are specified in Paper C. A decade-scale null closes the channel at its stated strength.

(iv) Population statistics over single objects. For stellivores and attenuated aestivation, the executable tests are statistical: regulation anomalies across binary catalogues; suppression statistics across old systems. These reuse public archives, cost salaries only, and would give the family's two least-tested members their first empirical constraints.

The program has a unifying property: every test doubles as conventional astrophysics (IMBH demographics, accretion physics, binary populations, cluster dynamics), so the field can pursue the family's adjudication without betting careers on its truth. The pattern is inherited from the waste-heat surveys (Wright et al., 2014b; Griffith et al., 2015) and from the falsificationist turn of Sandberg et al. (2016) and Dvali and Osmanov (2023), and it does not depend on the fate of any individual hypothesis.

8 Conclusion

Reviewed as a family, the inward hypotheses form a six-decade research lineage with a common physical premise, the thermodynamics of computation as the lens for predicting mature intelligence, and genuine internal disagreements about timing, destination, and visibility that observation can, increasingly, adjudicate. The family has moved from direction without destination (Ćirković & Bradbury), to a destination whose stated kill conditions outrun any funded instrument (transcension), to testability gained under critique (aestivation and its Bennett–Hanson–Riedel correction). The current generation states explicit targets, thresholds, and kill criteria. Each advance was forced by the previous generation's critics.

Against the sociological silence-solutions, the family's distinguishing asset is residue: physics leaves traces where policy does not. Its distinguishing liability is the optimization premise, which remains unargued at the selection-theoretic level where it would need to be argued (open problem P1). The practical situation of 2026 is: for the first time, members of this family have pre-registered tests scheduled on funded instruments, anchored by the globular-cluster surveys now beginning and by LISA's spin measurements a decade out. C5 already carries two partial credits (BH computing, MTH; Table 3), but no member has yet scored a full grade there, since no pre-registered inward-family test has been executed to completion. Within fifteen years that should change, for a literature long accused of unfalsifiability.

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Revision record

v2.6 (2026-10-22). The thermodynamic objection to aestivation was re-attributed: Bennett et al. (2019) argue that erasure cost is not lowered by waiting, and the harvest-timing argument (free energy not collected promptly is lost) is stated as this review's own. The Ren et al. contamination count was corrected from "four more" to "five more" (two significant, one suggestive, four marginal; two of nine without an identified contaminant). Bibliographic corrections: the Carroll-Nellenback DOI, the Carrigan name order, missing DOIs for Ren (2025), Huang (2026) and the Penrose reprint, volume and page data for Chen (2026), Whitaker (2026) and Kuzma (2026). The scorecard robustness file `figs/fB_calc7_scorecard.json` was regenerated against the v2.5 table; the totals for MTH under the five codings are 4.5, 3.0, 6.0, 4.5, 4.0.

Data availability

No new observational data were generated for this work. All quantitative claims derive from the cited literature.

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